

Skill checklist — protstruct-eval applied end-to-end

All 8 steps green (step 8 new in this re-issue) · plus water / ligand / metal sub-checklist

#	Step	Result
1	Re-derive R-factors with phenix.model_vs_data	agent 0.149 / 0.199 · oracle 0.156 / 0.211
2	Re-extract atom + water counts	atoms 1641 vs 1654 · waters 146 vs 159 · both Δ −13
3	Re-run peak finder at 4 σ	23 + / 8 − · agent listed 9 · max +6.18 σ
4	Verify quoted coordinates	Ca ²⁺ position 0.215 Å off (initial placement, not final)
5	Anomalous-data check	none in 1sar.mtz · closed by CheckMyMetal-style geometry
6	Per-round Δ -claim	round 6 R-free 0.207 < round 7 0.213
7	Non-cctbx confirmation per finding	8 / 10 oracles tagged · cross_tool_coverage all closed
8	T13 data-quality oracle (NEW)	ctruncate: Wilson B 14.5 · L-test 0.03 · Δ B 7.3 · no tNCS · 1 borderline ice ring

Water / ligand / metal sub-checklist

Check	Done	Result
Per-ion RSCC	✓	Ca ²⁺ 0.972 · Na ⁺ 0.964 · SO4 0.988 — all > 0.85
Per-ion B-vs-protein	✓	Ca ²⁺ 2.67× · Na ⁺ 1.43× · SO4 1.10×
Coordination geometry	✓	CN = 3 inner-sphere both ions (low — partial-occ framing)
Element identity (geometry)	✓	CheckMyMetal-style: Mg ²⁺ ruled out by IZI 14–16
Element identity (anomalous)	n/a	no anomalous columns in 1sar.mtz
Per-water B + RSCC distribution	✓	mean B 20.76 · mean RSCC 0.867 · 3 waters RSCC < 0.7
Density-misfit ResidueOutliers	✓	HOH S 680, 707, 729